Sri Neeraj Reddy Palakolanu

+91 8341677532 srineerajreddyp@gmail.com Sri Neeraj Reddy Sri Neeraj Reddy Portfolio

Education

•Vellore Institute of Technology *Aug. 2022 – Jun 2026*
Computer Science and Engineering, B.Tech. Amaravati, India
– **CGPA**: 8.8

•Narayana Junior College *Aug. 2020 – May 2022*
Science MPC Vijayawada, AP, India
– **Percentage**: 92%

Projects

•PingDrop – Android & Mac Companion App *Mar 2026 – Present*
Kotlin, Jetpack Compose, Swift, SwiftUI, Bluetooth LE, Wi-Fi TCP, AES-256-GCM **Google Play** | **Website**

– Built and shipped a free Android + macOS app (Google Play and a notarised Mac app) that links a phone and a Mac with no account or cloud — direct, encrypted Wi-Fi and Bluetooth LE with automatic handover.
– Notification mirroring with replies, calls (including WhatsApp), file and folder transfer, clipboard sync, phone as a Mac webcam, walk-away auto-lock, media controls, widgets and Siri Shortcuts.
– Ran it end to end: Play Console releases and in-app billing (Pro tier), notarised Mac builds with auto-updates, the website and launch marketing.

•PolicyMind AI – Enterprise Policy Assistant *Aug 2026*
Python, RAG, ChromaDB, Gemini, Streamlit, Sentence Transformers **Project Repository**

– Built a multi-document RAG system that processes company policy PDFs, performs semantic and keyword-based retrieval, and generates grounded answers using Google Gemini.
– Implemented persistent ChromaDB vector storage with document metadata, page-level source citations, and automated PDF indexing.
– Improved retrieval quality using hybrid search with BM25 and semantic retrieval, followed by cross-encoder reranking, relevance filtering, and deduplication.
– Developed a Streamlit interface with conversation memory and follow-up question rewriting for context-aware document querying.

•Efficient ANPR – Automatic Number Plate Recognition *Jan 2026*
Python, YOLOv8n, DeepSORT, PaddleOCR, OpenCV, Ultralytics, Google-Colab, Kaggle **Project Repository**

– Trained two YOLOv8n models on Indian traffic datasets, one for vehicle detection (21000+ images) and one for number plate (2000+ images) recognition.
– Built an end-to-end video inference pipeline passing vehicle bounding boxes into the plate model, then cropping plate regions for OCR using PaddleOCR.
– Integrated DeepSORT tracking to assign persistent IDs to vehicles, ensuring each vehicle undergoes OCR only once, improving pipeline efficiency and computation.

Skills

- **Programming Language**: Python, Java, SQL
- **ML / AI**: Deep Learning, LLMs, RAG, YOLO, OpenCV
- **Databases**: MySQL, ChromaDB
- **Tools**: Git, Visual Studio Code, Jupyter Notebook, GitHub, Google Colab

Positions of Responsibility

•Documentation Team Lead, LEO Club, VIT - AP *Nov 2024 – May 2025*

•Core Team Member, Machine Learning Club, VIT - AP *Aug 2023 – Dec 2023*

•Event Management Team Member, NGC Club, VIT - AP *Jan 2024 – May 2024*

Licences & Certifications

•AWS Certified Cloud Practitioner, Amazon Web Services *2025*

•PCAP – Programming Essentials in Python, Python Institute *2024*